RAMIRO YAIR ROMÁN TREJO

Junior Backend Developer (Python / Django)

Zacatecas, Mexico • (+52) 492-181-9531 • ryrt32@gmail.com • roman-mx.tech • github.com/YairRT45 • linkedin.com/in/ramiro-yair-román-trejo

WORK EXPERIENCE

Software Developer Intern — Compañía Cervecera de Zacatecas (Grupo Modelo / AB InBev) Aug 2026 – Oct 2026

- Digitized operational audits and tool tracking in the Packaging Department, replacing manual paper logs with a centralized **internal web application**.
- Built the backend and **relational database (SQL)** using **Python (Django)** and **PostgreSQL**, automating issue tracking and corrective action workflows on the plant floor.
- Deployed the application in **Docker** containers and connected operational data to **KPI dashboards in Power BI** for maintenance reporting.

Quality Assurance (QA) Inspector — Nagakura Engineering Works Apr 2021 – Jul 2022

- Executed quality control and dimensional part inspections across automotive manufacturing lines under strict international production standards.
- Documented defects and collaborated with the engineering team on **root cause analysis** to implement corrective actions and reduce line rejections.

SOFTWARE DEVELOPMENT PROJECTS

Academic Curriculum Tracking Platform (UAZ – Lic. en Lenguas Extranjeras) | Python, Django, MySQL/MariaDB, AWS, Docker Nov 2025 – May 2026

- Developed the backend for an institutional web platform using **Python (Django)** and **MariaDB (MySQL)** to manage course syllabi and faculty tracking, replacing manual spreadsheets.
- Configured automated deployment (**CI/CD**) to a **Linux cloud server (AWS)** using **Jenkins** and **Docker**, integrating **automated QA tests** with **Selenium** and **Behave**.

Crop Pathology Detection System (Doctor Chile) | Python, PyTorch, Flutter, SQLite github.com/YairRT45/doctor_chile_app

- Trained a **Computer Vision / AI model (Python, PyTorch)** to detect 5 foliar pests and diseases in chili crops from field photographs.
- Built a mobile application in **Flutter** with a local **SQLite** database, enabling farmers to run diagnostics and track crop history offline in rural areas.

University Enrollment & Scheduling System (UPN-SASCE) | Python, Django, PostgreSQL, Docker Institutional Project

- Developed student enrollment and class scheduling modules for **Universidad Pedagógica Nacional (UPN)**, validating course prerequisites and capacity rules through **automated testing**.

Data Structures & Graph Algorithms in Java | Java github.com/YairRT45

- Implemented core data structures from scratch (Binary Trees, Heaps, Graphs) and shortest-path routing algorithms in **Java**, along with a stack-based syntax parser.

TECHNICAL SKILLS

Languages & Backend: Python, Django, RESTful APIs (Django REST Framework, FastAPI), SQL, Java, HTML / CSS.

Databases, Servers & Cloud: Relational Databases (PostgreSQL, MySQL / MariaDB, SQLite), Web Servers (Apache, Nginx), Linux, AWS (EC2), Docker.

Tools, CI/CD & QA Testing: Git, GitHub / GitLab, CI/CD Pipelines (Jenkins, GitHub Actions), Automated Testing (Selenium, pytest, Behave), Agile / Scrum.

EDUCATION & LANGUAGES

B.S. in Software Engineering — Universidad Autónoma de Zacatecas (UAZ) 2022 – Present (Final Semester)

Languages: Spanish (Native) • English (B1 – Intermediate: technical documentation reading and written communication).